

Neural plasticity

- Most of our knowledge and abilities are stored in neural parameters
 - above all, in synapses

• Can this information be stored in genes? Quick estimate:

Ca. 10^{15} synapses in (human) cortex

Ca. 10^{10} base pairs à 2 bit each

$$\hookrightarrow \frac{10^{10} \cdot 2}{8 \cdot 2^{30}} \approx 2 \text{ GB}$$

→ Syn. configuration cannot
be encoded in genes

- Solution: Neural plasticity and experience-dependent learning

• Typical dimensions of plasticity:

- Neurons: shape: dendritic tree, axon growth, ...

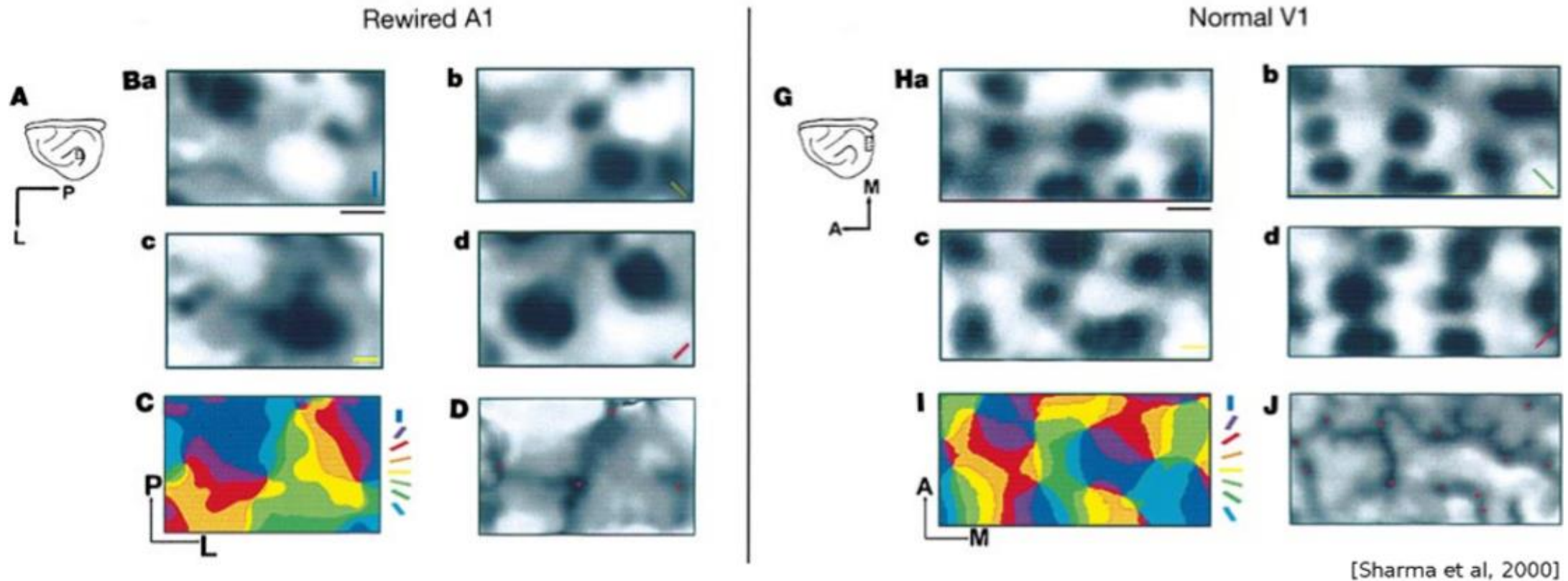
parameters: ion channel density, conductance, ...

- Synapses: structural plasticity: location/existence of syn.

syn. weights: size, transmitter release probability, ...

→ young ferrets
(Sherman et al.)

- Examples of plasticity in neural networks: Rerouting of visual fibers to auditory cortex



Auditory fibers were rerouted to visual cortex V1 at very early stage of cortical development.

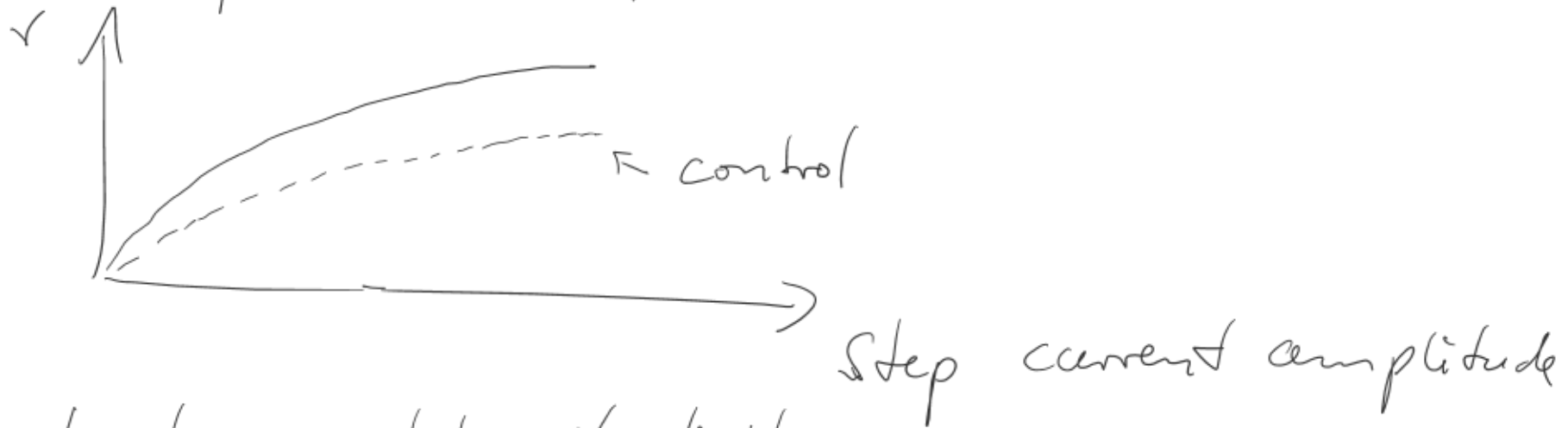
Normal path: retina -> visual thalamus -> V1

Experimental modified path: retina -> auditory thalamus -> A1

Result: Formation of orientation tuned clusters in A1, similar to V1 -> functional reorganization due to plasticity

Intrinsic plasticity

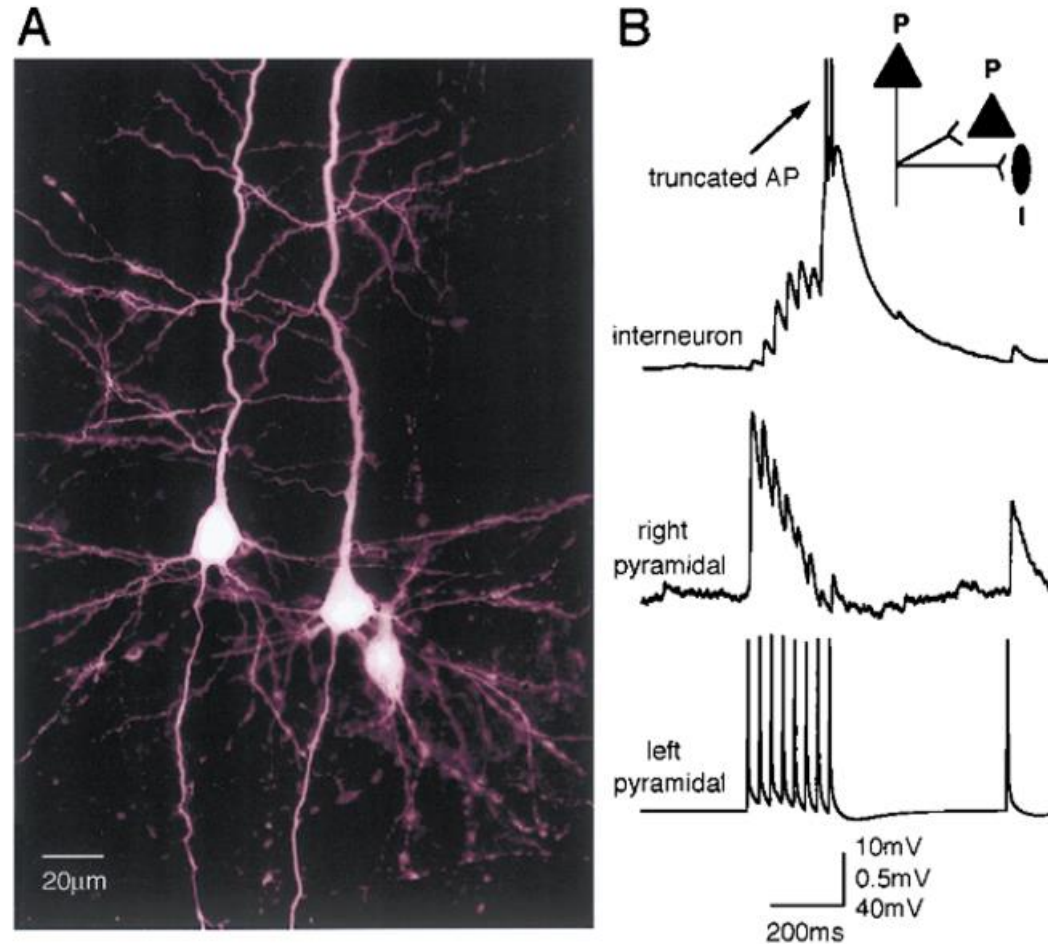
- The excitability of neurons changes based on past stimuli (mainly by regulating ion-channel conductances).
- Example: stronger response after activity deprivation



↳ homeostatic plasticity

- But: Also the opposite is observed (e.g. increased excitability after strong stimuli)

Short term plasticity (STP):
Facilitation and depression

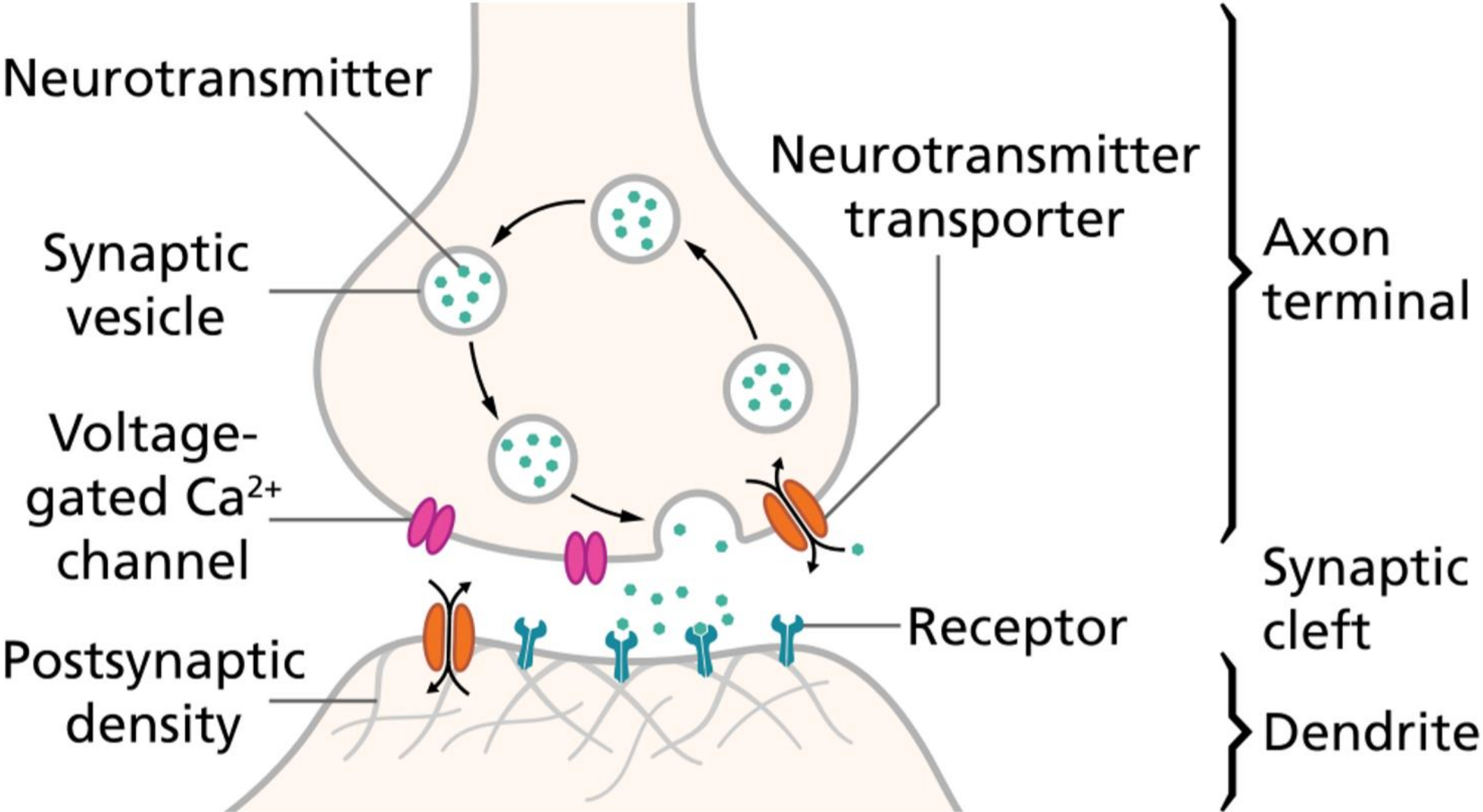


facilitation in interneuron
and depression in right
pyramidal

FIG. 2. Differential synaptic facilitation and depression via the same axon innervating two different targets. (A) A light microscopic pseudocolor image of three biocytin-filled neurons. The pyramidal neuron on the left innervated the pyramidal neuron on the right and the bipolar interneuron on the right. (B) Single trial responses (30 Hz) to same AP train. Failure rate for first EPSP: interneurons, 24%;

[Markram et al, 1998]

Understand the cycle of neurotransmitters for the STP model



Short-term synaptic plasticity (STP)

- Synapses change their efficacy on the time scale of tens of ms to seconds ... and recover again.

The Trodyus-Markram (TSM) mechanism

- Phenomenological model, but inspired by biophysics
- Three partitions (= possible states of neurotransmitter)



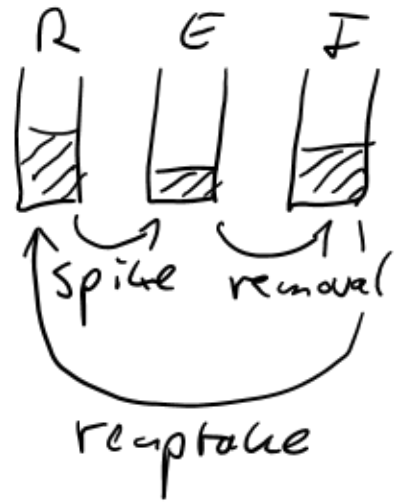
- R: Recovered
 - E: Effective
 - I: Inactive
- } fraction of transmitter
 $\hookrightarrow R + E + I = 1$

$\hookrightarrow$ Synaptic conductance $g^{\text{syn}}(t) \propto E(t)$

$\hookrightarrow$ The fraction of R released at action-potential at time t_s is called "Utilization" U : $\Delta E = U \cdot R(t)$ at spike-time t_s

$\hookrightarrow$ "Ground state" (after long quiescence): $(R, E, I)_0 = (1, 0, 0)$.

• (Formal) synapse dynamics



• Pre-synaptic spikes $g(t) = \sum_s \delta(t - t_s)$

(1) Transmitter release at spike

$$\frac{\partial R}{\partial t} = -g(t) \cdot U \cdot R(t) + \frac{I(t)}{\tau_{rec}}$$

(2) Transmitter removal

$$\frac{\partial E}{\partial t} = +g(t) \cdot U \cdot R(t) - \frac{E}{\tau_{inact}}$$

leave blank initially

(3) Transmitter reuptake

$$\frac{\partial I}{\partial t} = +\frac{E}{\tau_{inact}} - \frac{I}{\tau_{rec}}$$

($\tau_{inact} \approx \tau_{syn}$)

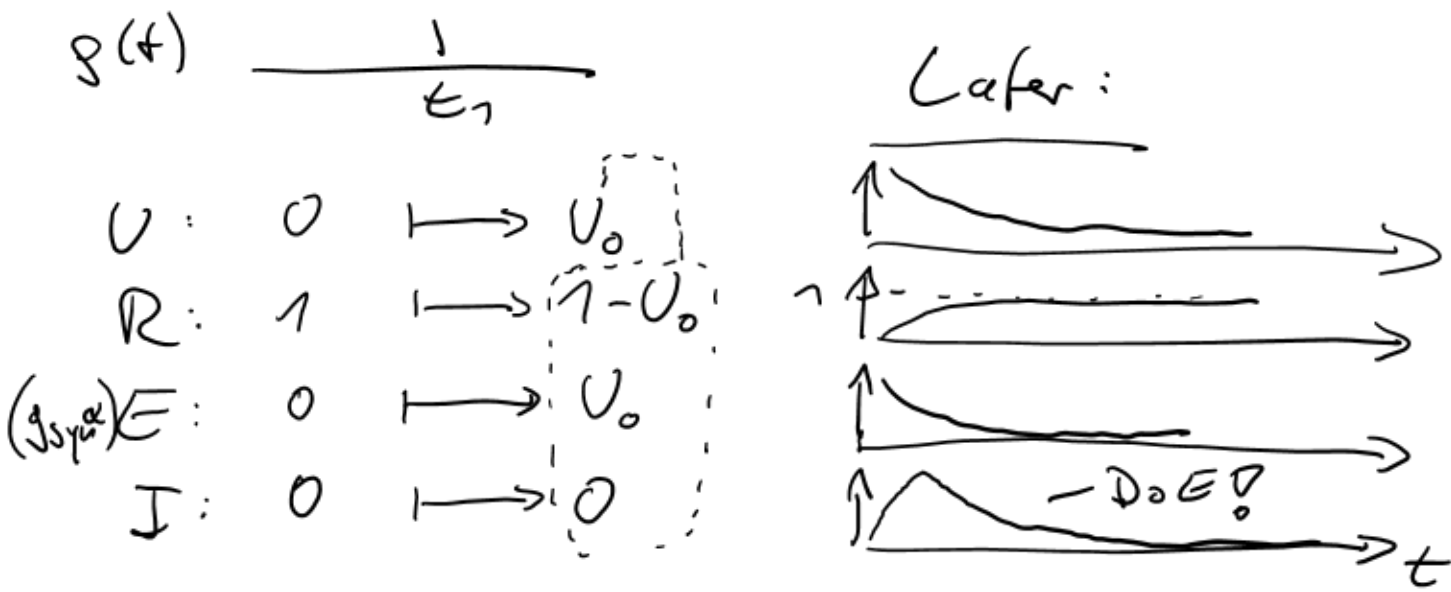
$$= -\left(\frac{\partial E}{\partial t} + \frac{\partial R}{\partial t}\right) \text{ since } R + E + I = 1$$

• For $U = U_0 = \text{const}$, this leads to vesicle depletion in case of high pre-syn firing rates
→ short-term depression

• Include facilitation via an adaptive utilization variable $U(t)$, $0 < U < 1$:

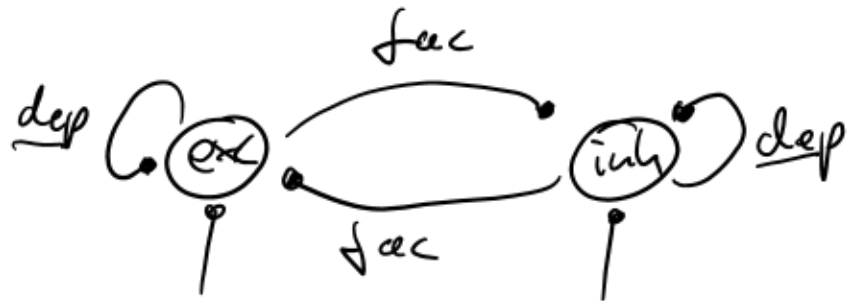
$$\frac{\partial U}{\partial t} = -\frac{U}{\tau_{fac}} + g(t) \underbrace{U_0(1-U)}_{\text{change towards 1.}}$$

Example: 1st spike at t_1 after quiescence

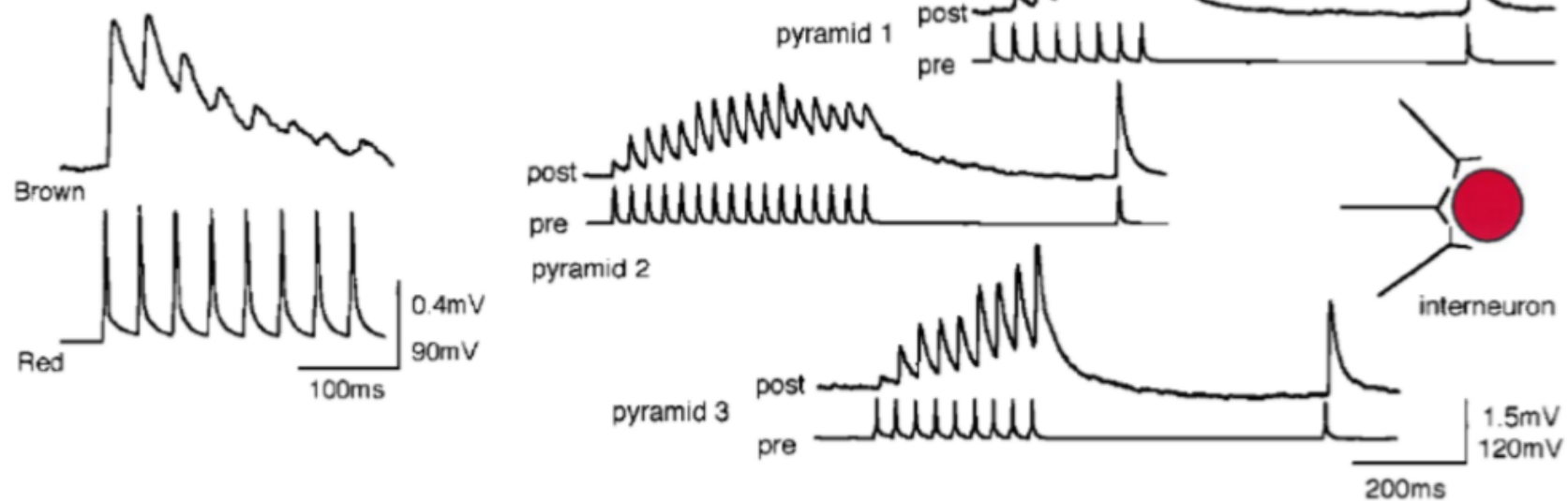


Network-level application

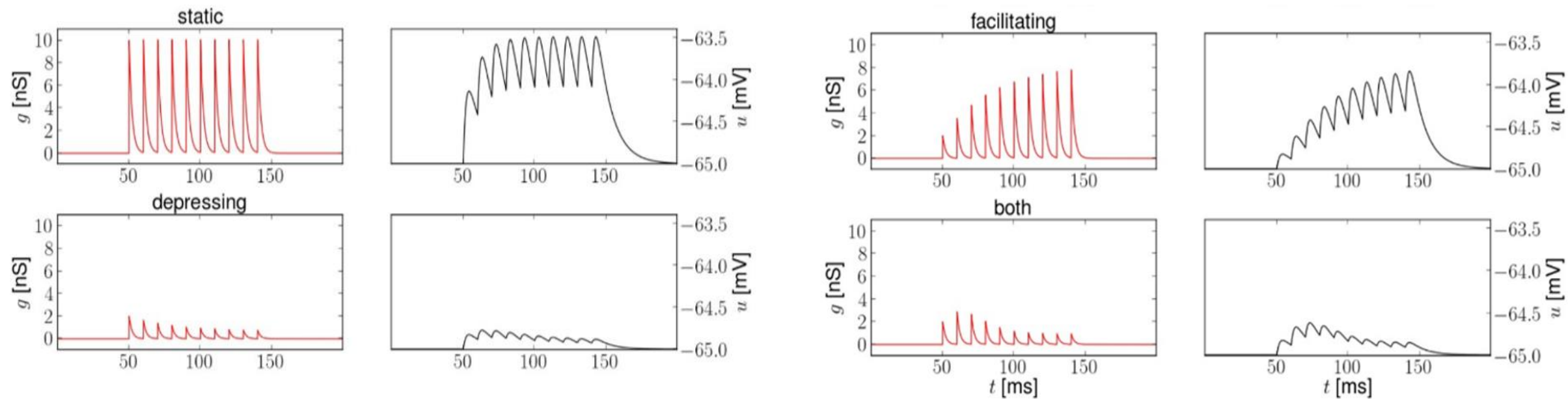
- Self-stabilization of recurrent networks via STP



↳ The network activity maintains a moderate firing rate.



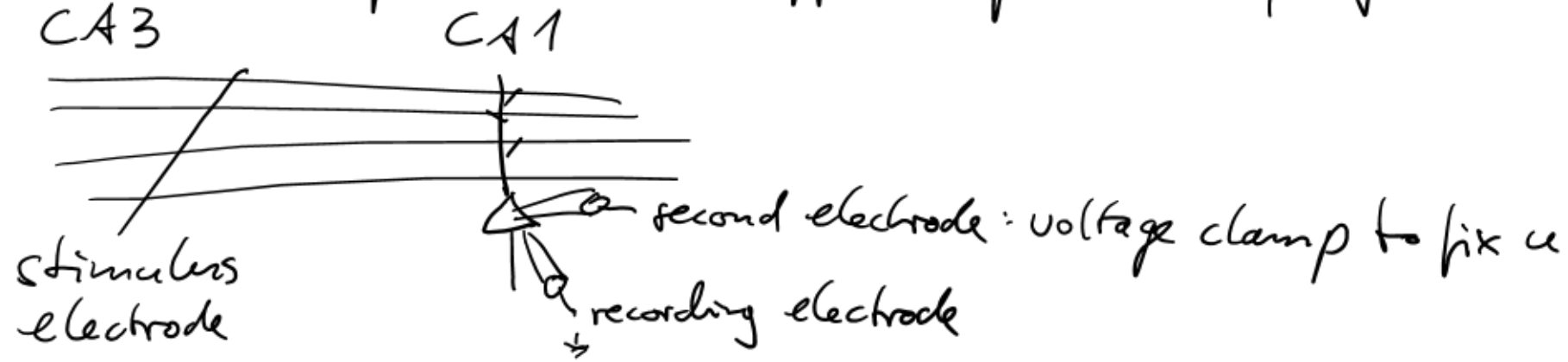
[Markram et al, 1998]



[Simulation by M. Petrovici]

Long-term synaptic plasticity

- classical experiment in hippocampus (very organized connectivity structure)



- After strong stimulation: Increased EPSP amplitude that remains stable over minutes to hours. $\rightarrow$ long-term potentiation (LTP)
- After weak stimulation: Decreased EPSP amplitude.
 $\rightarrow$ Long-term depression (LTD)
- 2nd experiment: clamp the post-synaptic neuron to some fixed potential
 $\hookrightarrow$ Voltage-dependent transition from LTD to LTP.

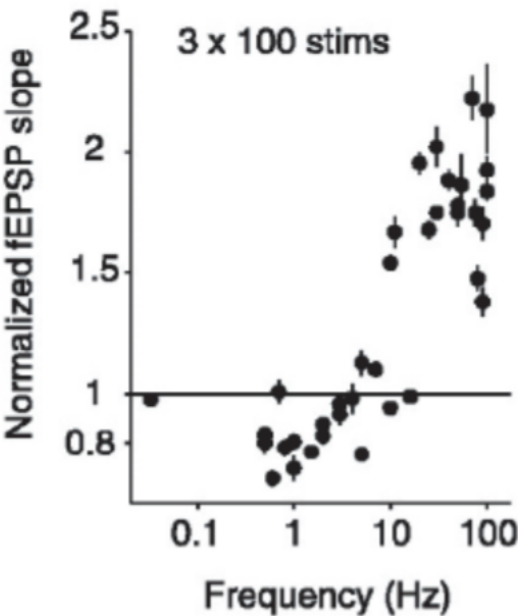
Rate-based experiments

High-frequency stimulation (LTP)

Presynaptic stimulation: [stimulation pattern] 100 Hz, 1 s
Postsynaptic activity: not controlled, not measured

Low-frequency stimulation (LTD)

Presynaptic stimulation: [stimulation pattern] 1 Hz, 900 s
Postsynaptic activity: not controlled, not measured



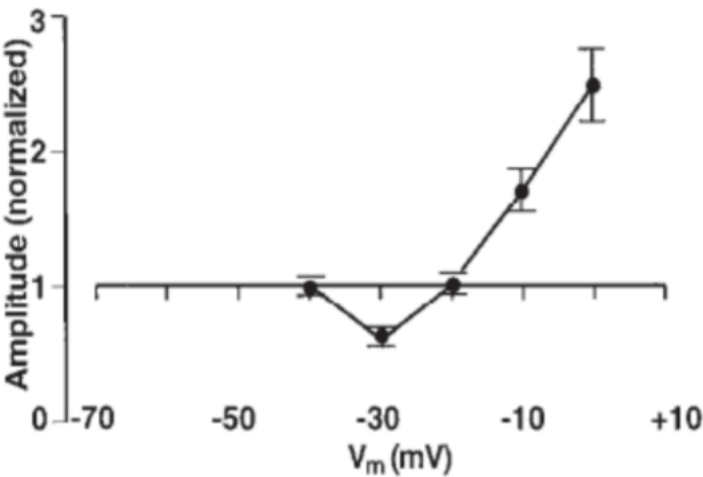
Studying the effect of postsynaptic potential through clamping (keeping the potential constant)

Strong depolarization (LTP)

Presynaptic stimulation: [stimulation pattern] 1 Hz, 100 s
Postsynaptic activity: [clamping pattern] 0 mV, -70 mV

Weak depolarization (LTD)

Presynaptic stimulation: [stimulation pattern] 1 Hz, 100 s
Postsynaptic activity: [clamping pattern] -30 mV, -70 mV



What happens in between?

Single cell experiments

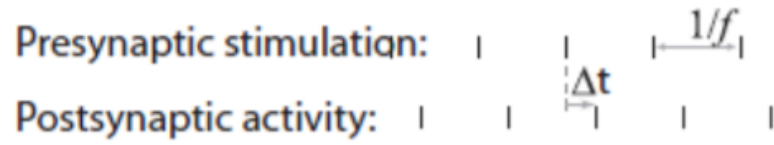
- The precise relative timing of pre- and post-synaptic spikes becomes important:



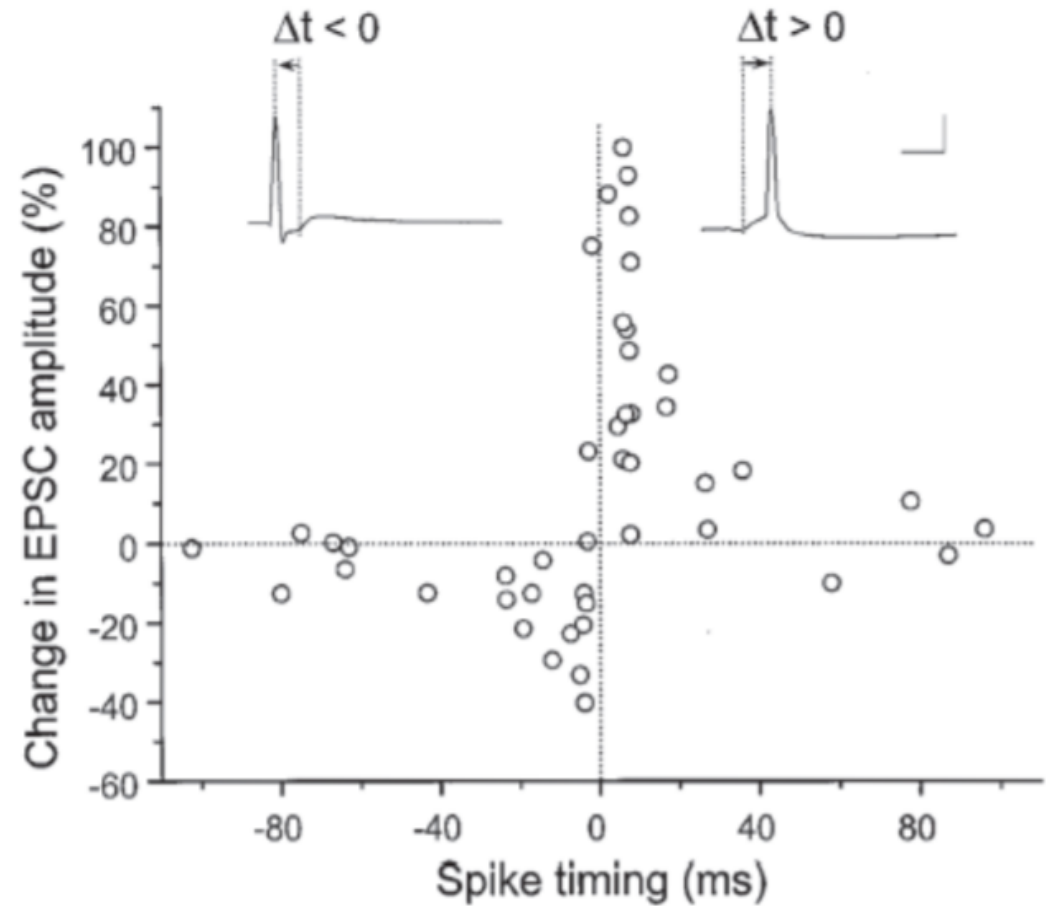
[Bi & Poo, 1998]

- This is known as: Spike-timing dependent plasticity (STDP)
- In this (classical) form, causal spike pairs ("pre-before-post") induce LTP, while acausal spike pairs ("post-before-pre") induce LTD.

Timed-spike stimulation



With the advancement of single-cell measurements, such experiments at the cellular level became possible, which led to these discoveries.



The previous experiments agreed with Hebb's postulates

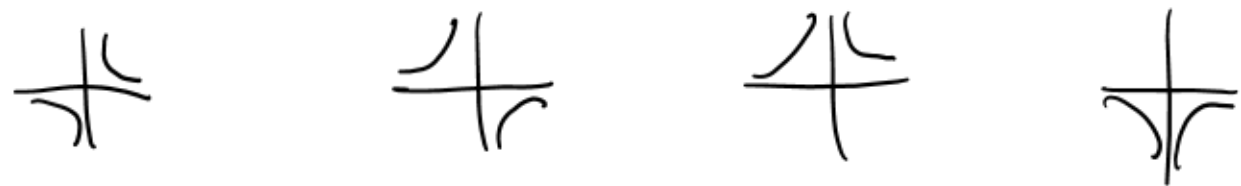
When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased."

Donald O. Hebb, 1949

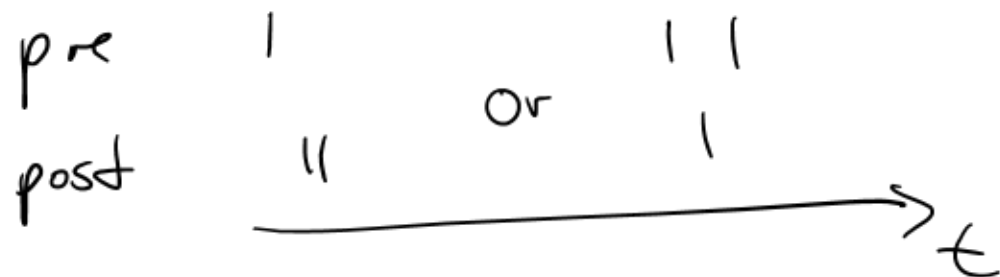
- This was postulated by Donald Hebb in 1949! → "Hebb's rule"

A word of warning:

- STDP is not uniform and universal!
- Various STDP curves have been observed:



- Plasticity is non-linear for, e.g. triplets



→ not: $\sum \text{spike-pairs}$

- Furthermore: neuron-, weight-, location-, voltage-, age-, neuro-modulator dependent, ...

Biophysical foundations of long-term synaptic plasticity

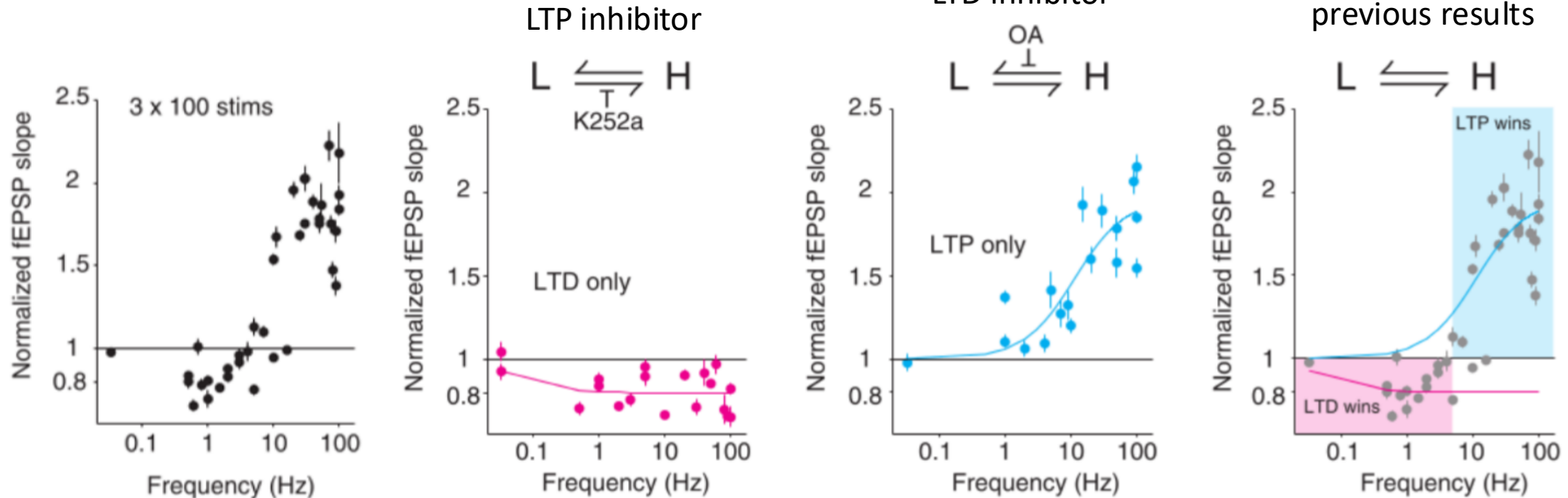
A general dilemma

In synaptic plasticity, diverse and complex bio-chemical pathways are involved at every synapse, leading to a confusing situation. At the same time, simplified plasticity models can easily generate a false and „dangerous“ picture.

The following are some examples of fundamental biophysical processes, which might all be simultaneously present in a synapse!

Separating LTD and LTP

Is LTD just reverted/negative LTP?



O'Connor et al, Dissection of Bidirectional Synaptic Plasticity Into Saturable Unidirectional Processes, J Neurophysiol (2005)

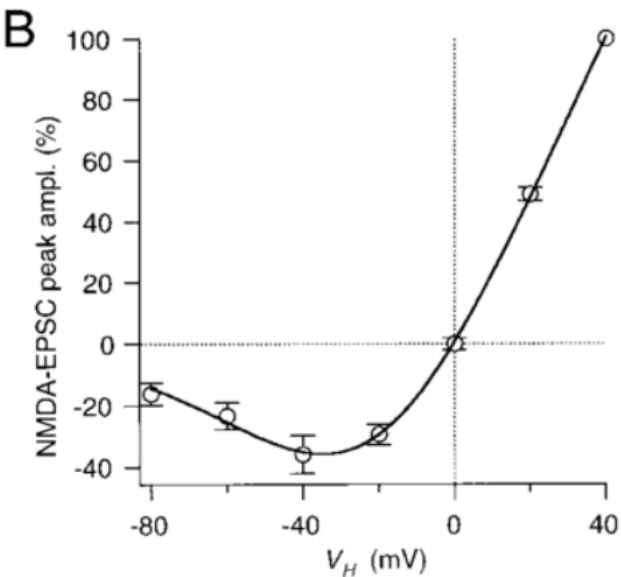
→ LTD and LTP are (at least to some extent) implemented by different bio-chemical pathways.

The specific role of NMDA receptors (NMDAR)

NMDARs control Ca^{2+} influx AND NMDARs act as coincidence detectors of

- 1) presynaptic neurotransmitter release (glutamate)
- 2) post-synaptic membrane potential (voltage-dependence)

The resulting $[\text{Ca}^{2+}]$ concentration triggers, e.g., the insertion or synthesis of new AMPA receptors



Voltage-dependent conductance of the NMDAR
(DOI: 10.1152/jn.2001.85.1.32)

Some important synapse types

Receptor	Neurotransmitter	Ion types	E_rev	τ_open	Comments
AMPA	Glutamate	Na ⁺ , K ⁺ , (Ca ²⁺)	ca. 0 mV	ca. 1-2 ms	Prevalent excitatory synapse
NMDA	Glutamate	Na ⁺ , K ⁺ , Ca ²⁺	ca. 0 mV	ca. 20 ms	+ voltage dependence
GABA_A	GABA	Cl ⁻	ca. -70 mV	ca. 5 ms	Prevalent inhibitory synapse
Ach	Acetylcholine	Na ⁺ , K ⁺		ca. 1 ms	Motor neuron → muscle contraction

AMPA: α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid
NMDA: N-methyl-D-aspartate
GABA: gamma-aminobutyric acid

Examples of molecular plasticity mechanisms

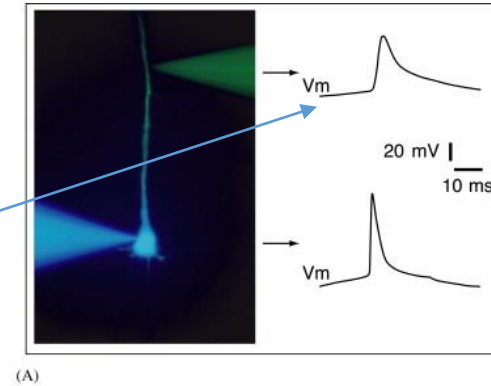
1. Post-synaptic LTP

Post-synaptic NMDARs are activated by

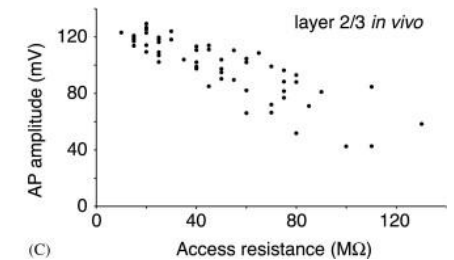
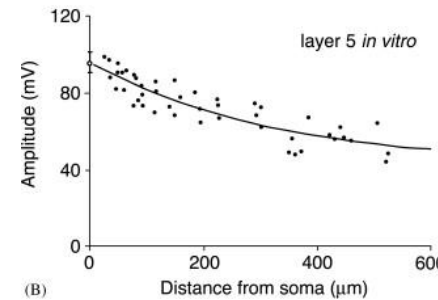
- 1) pre-syn glutamate release
 - 2) a high post-syn potential at the dendrite
- Factor 2) could be realized by back-propagating action potentials (bAP) after somatic spiking of the post-syn. neuron

2. Post-synaptic LTD

Without the bAP, the Ca^{2+} influx will be smaller and often triggers only an LTD pathway.



Amplitude of the bAP depending on the distance from the soma on the long (apical) dendrite



3. Pre-Synaptic LTD

Coincidence of two conditions:

- 1) NMDA auto-receptors are „automatically“ activated upon the pre-syn. AP and Glutamate release.
- 2) Retrograde signalling of endo-cannabinoids (CB) released from the post-syn. Membrane and „detected“ by CB1 receptors. The CB concentration (seems to) depend on the recent post-syn. avg. membrane potential.

→ Triggers changes in pre-synaptic transmitter release.

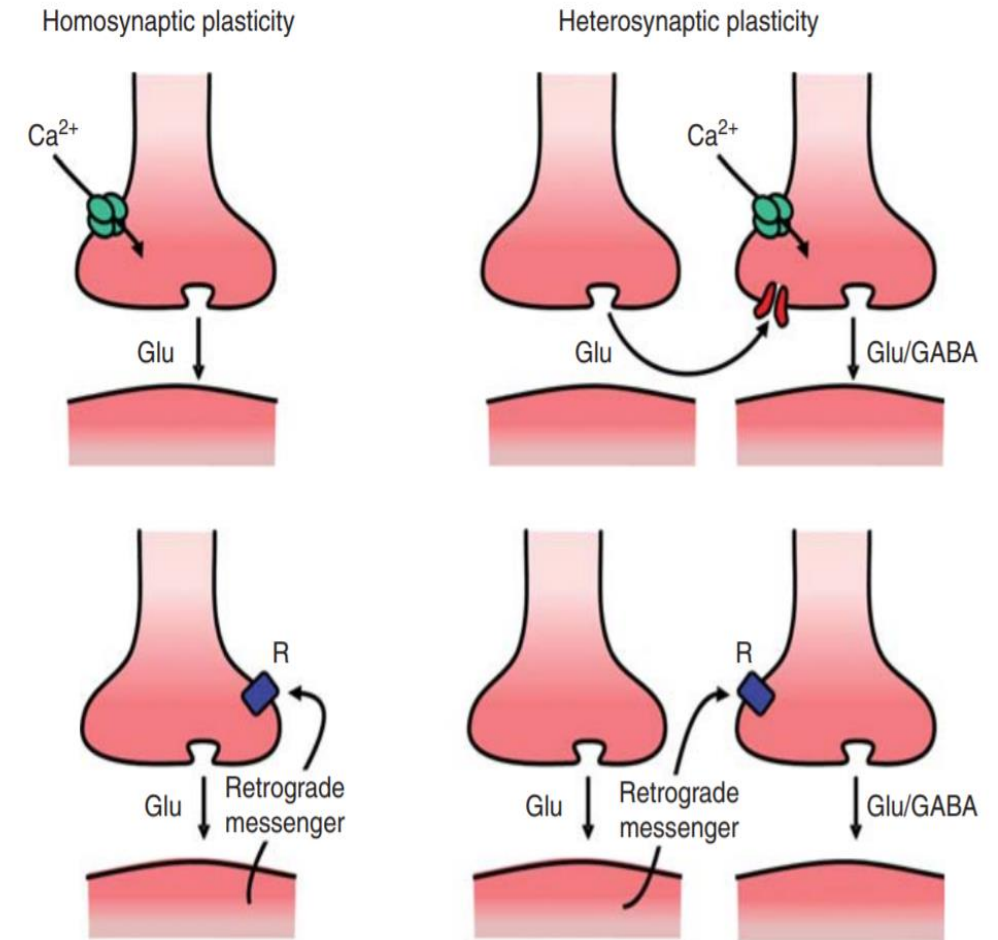


Figure 1. Four basic arrangements for presynaptic LTP and LTD. (*Top row*) Illustrates examples where plasticity is entirely induced presynaptically, either in a homosynaptically (e.g., mossy fiber LTP) or heterosynaptically (e.g., associative cortico-lateral amygdala LTP of glutamatergic transmission). (*Bottom row*) Examples requiring retrograde signaling generated either at the same synapse receiving the signal (*left*, e.g., eCB-LTD at excitatory synapses) or at a neighboring synapse (*right*, e.g., eCB-LTD at inhibitory synapses).